

Order Flow Imbalance and Price Impact in CME Ether Futures

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Abstract

We study Order Flow Imbalance (Cont, Kukanov, and Stoikov, 2014) on CME Ether futures over February 2021 to August 2026. On 28,711 out-of-sample dollar-volume bars, an OLS regression of bar mid-quote change on contemporaneous OFI with Newey-West HAC standard errors yields $\hat{\beta}_0 = +0.215$ ($t = 58.1$, $R^2 = 0.30$), against $+0.300$ and $R^2 = 0.31$ in sample. The regression is contemporaneous, so it measures price impact rather than forecasting power: used on its own as a predictor, lagged OFI explains essentially none of the following bar’s move ($R^2 < 0.001$ out of sample). OFI is a strong descriptor of price formation on this contract; it is not, by itself, a trading signal.

1 Introduction

Order Flow Imbalance (OFI) aggregates best-bid and best-offer quote arrivals into a signed measure of top-of-book pressure. On liquid US equities, contemporaneous OFI explains roughly two-thirds of bar-level mid-quote variation (Cont, Kukanov, and Stoikov, 2014). The cryptocurrency evidence is thinner: Silantsev (2019) finds trade-flow imbalance dominates short-horizon prices on BitMEX XBT/USD, and Bieganowski and Ślepaczuk (2026) report OFI as the top SHAP feature in CatBoost ensembles on Binance Futures perpetuals. CME’s Ether contract sits apart from both: a regulated venue whose quoted spread, in basis points of price, is an order of magnitude wider than the equity tape’s.

This paper estimates the contemporaneous OFI signal-impact on CME ETH futures from February 2021 to August 2026. The construction is the level-1 (best bid / best offer) form of Cont, Kukanov, and Stoikov on dollar-volume bars built from TBBO (trade-sampled top-of-book) events (Lopez de Prado, 2018).

2 Data

The panel consists of CME-cleared TBBO data for the front-month ETH futures contract, covering February 2021 through August 2026, with rolls handled by contract stitching.

Bars are constructed in dollar volume, with the threshold calibrated so that a median in-sample trading day produces about ten bars, and that threshold is held fixed when bars are constructed for the out-of-sample period.

For each bar the following quantities are computed:

- **ofi**: signed level-1 (best bid / best offer) order flow imbalance, following the construction of Cont, Kukanov, and Stoikov (2014). For each TBBO update i , let p_i^B, p_i^A denote the

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best-bid and best-ask quotes and q_i^B, q_i^A their sizes. The per-event OFI contribution is $e_i = e_i^B + e_i^A$, where

$$e_i^B = \begin{cases} +q_i^B & \text{if } p_i^B > p_{i-1}^B \\ -q_{i-1}^B & \text{if } p_i^B < p_{i-1}^B \\ q_i^B - q_{i-1}^B & \text{if } p_i^B = p_{i-1}^B \end{cases}, \quad e_i^A = \begin{cases} -q_i^A & \text{if } p_i^A < p_{i-1}^A \\ +q_{i-1}^A & \text{if } p_i^A > p_{i-1}^A \\ -(q_i^A - q_{i-1}^A) & \text{if } p_i^A = p_{i-1}^A. \end{cases}$$

The bar OFI is the unnormalised sum over all events in the bar, $O_t = \sum_{i \in \text{bar } t} e_i$.

- **mid_close**: the mid quote $(p^B + p^A)/2$ at the bar boundary.
- **d_mid**: the change in mid quote since the previous bar.

3 Signal-impact regression

Following Cont, Kukanov, and Stoikov (2014) and Silantyev (2019), bar mid-quote change is regressed on contemporaneous and one-bar lagged OFI:

$$\Delta p_t = \alpha + \beta_0 O_t + \beta_1 O_{t-1} + \varepsilon_t, \quad (1)$$

where $\Delta p_t = m_t - m_{t-1}$ is the change in mid-quote between bar boundaries and O_t is the bar OFI, so β_0 is in dollars of mid-quote change per contract of net imbalance. Standard errors are Newey-West HAC (Newey and West, 1987) with five lags. Table 1 reports the estimates on the full panel, the 2021-02 to 2023-12 in-sample (IS) window, and the 2024-01 to 2026-08 out-of-sample (OOS) window.

Window	n (bars)	$\hat{\beta}_0$	HAC $t(\beta_0)$	$\hat{\beta}_1$	HAC $t(\beta_1)$	R^2
Full sample	41,055	+0.247	+70.2	-0.027	-14.4	0.29
In-sample 2021–23	12,344	+0.300	+44.9	-0.025	-7.0	0.31
Out-of-sample 2024–26	28,711	+0.215	+58.1	-0.026	-13.3	0.30

Table 1: OLS, Newey-West HAC (five lags). $p < 10^{-6}$ on both coefficients in every window.

Both subsamples reject $\beta_0 = 0$ at $p < 10^{-6}$ (Figure 1). The IS and OOS R^2 are nearly identical (0.31 and 0.30), while the slope itself is less stable, moving from +0.300 in-sample to +0.215 out-of-sample. Both R^2 figures are below the two-thirds level of Cont et al. (2014) for liquid US equities.

This regression is contemporaneous: it measures how the mid quote moves together with the flow inside a bar (mean OOS bar duration ≈ 49 minutes), not whether flow forecasts the next bar. Used on its own as a predictor, lagged OFI explains essentially none of the following bar’s move ($R^2 < 0.001$ out-of-sample), so the content of Table 1 is price impact rather than forecastability. Whether the flow carries predictive content at horizons inside the bar is a question the bar regression cannot resolve, and answering it properly requires quote-level data.

4 Conclusion

The contemporaneous OFI signal-impact is strong and stable on CME Ether futures: the explanatory power is nearly identical in and out of sample (R^2 of 0.31 and 0.30), and the slope remains large and precisely estimated throughout. It is a statement about price impact, not forecastability; lagged OFI alone explains essentially none of the next bar’s move, so the result describes how this market absorbs order flow rather than offering a trading signal.

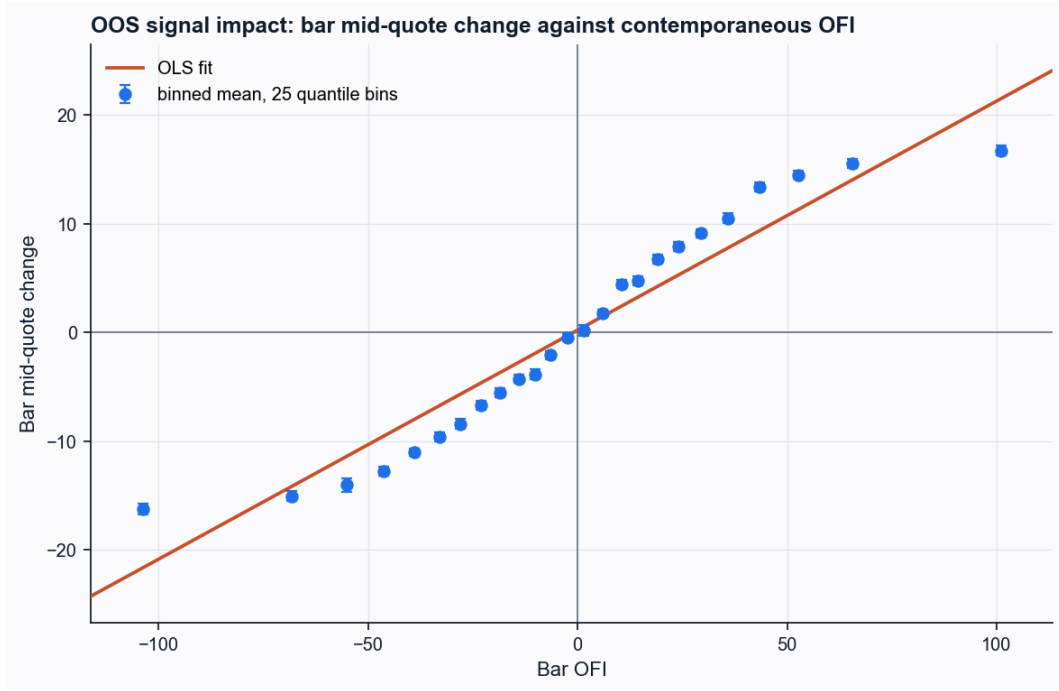


Figure 1: Out-of-sample bar mid-quote change against contemporaneous OFI, 25 quantile bins with ± 1 standard error and the OLS fit. The relationship is monotonic, with impact per unit of flow attenuating at extreme flows: both tail bins fall inside the straight-line fit. Full estimates are in Table 1.

The tape bounds what can be measured. TBBO is trade-sampled, so the book is observed only when a trade prints, and it carries no depth beyond the touch. A quote-level feed would permit measuring the relation at horizons inside the bar, where any predictive content would live, and a proper treatment of execution; these are the natural extensions.

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